

ML Lab Assignment 1

Write python code for the following and note down your observations and analysis.

```
function min_var_change_count( k=100, N){  
    Generate K arrays of size N each, using random number generator  
    For each array, find the minimum number in the array using following logic  
    temp_min = inf  
    var_change_count=0  
    i =0  
    while i < size of array :  
        if temp_min < array [i]  
            temp_min = array [i]  
            var_change_count ++  
        i++  
    store value of var_change_count for each array in a list  
    Plot the histogram of var_change_count distribution and calculate mean, median and  
    geometric mean of this distribution!  
    return mean_var_change_count  
}
```

Repeat above procedure for following values of N 10, 20, 50, 100, 200, 500, 1000, 2000, 5000, 10000, 20000, 50000, 100000, 200000, 500000, 1000000

Plot graph of N vs corresponding mean_var_change_count